

CHANGE REQUEST

CR-04

Client-Facing Showcase Enhancement,
Downloadable Profile, Automation
& Business Intelligence Module

Base SRS: Sulfri Trainer Portal — Personal Brand Edition v1.0
Amendment extends: CR-01 + CR-02 + CR-03

CR ID	CR-04
Type	Functional Enhancement — Multi-Module (Core Business Operations)
Priority	High
Target Version	v1.4 (Post-Approval Consolidation)
Base SRS	Sulfri Trainer Portal — Personal Brand Edition v1.0
Extends	CR-01 (Badge Wallet) + CR-02 (Skill Tree) + CR-03 (Event Lead Capture)
Prepared By	System Specification Draft
Status	Pending Approval

Module A

Brand Identity & Core Messaging

Module B

Client Engagement & CRM

Module C

Proposal Generation & Analytics

Module D

Automated Reporting & Insights

Module E

Advanced Analytics & AI

Module F

SEO, Performance & Marketing Automation

TABLE OF CONTENTS

1. Executive Summary
2. Alignment with Existing SRS & CRs
3. Scope of Change
 - 3.1 In Scope
 - 3.2 Out of Scope (Future Phase)
4. Module A — Branded PDF Profile Download
 - 4.1 Functional Requirements (FR-100 – FR-112)
5. Module B — Client Showcase Enhancements
 - 5.1 Functional Requirements (FR-113 – FR-124)
6. Module C — Intelligent Proposal Generator
 - 6.1 Functional Requirements (FR-125 – FR-138)
7. Module D — Automation, CRM & Communication Pipeline
 - 7.1 Functional Requirements (FR-139 – FR-155)
8. Module E — Analytics & Admin Business Intelligence
 - 8.1 Functional Requirements (FR-156 – FR-168)
9. Module F — SEO, Performance & Mobile UX
 - 9.1 Functional Requirements (FR-169 – FR-178)
10. Database Amendments
11. API Amendments
12. Non-Functional Requirements
13. Acceptance Criteria
14. Impact on Deployment
15. Version Control & CR Log
16. Strategic Value Statement

1. EXECUTIVE SUMMARY

This Change Request introduces six interconnected enhancement modules that collectively transform the Sulfri Trainer Portal from a verified credential showcase (achieved in CR-01 through CR-03) into a fully operational client-facing business platform. The objective is to enable Mohd Sulfri Mohd Harris to actively use the portal as a live sales, canvassing, and client engagement tool — not merely a passive profile page.

The six modules introduced in this CR are:

Module A	Branded PDF Profile Download Auto-generated, professionally designed downloadable trainer profile (1-page & full version).
Module B	Client Showcase Enhancements Testimonials, case studies, live counters, logo grid, and social proof layers.
Module C	Intelligent Proposal Generator Client-driven form that auto-generates a tailored training proposal PDF.
Module D	Automation & CRM Pipeline Lead capture → auto-email → CRM tracking → WhatsApp CTA → follow-up automation.
Module E	Analytics & Admin BI Dashboard Traffic analytics, download tracking, lead source analysis, and engagement reporting.
Module F	SEO, Performance & Mobile UX Google indexability, sticky CTAs, page speed optimisation, and structured data.

2. ALIGNMENT WITH EXISTING SRS & CRs

This CR is fully aligned with the architectural and content-management philosophy established in SRS v1.0 and extended across CR-01, CR-02, and CR-03. It does not alter core system architecture, authentication, deployment model, or user role structure.

DOES NOT introduce new user roles	✓
DOES NOT change authentication model (JWT/NextAuth)	✓
DOES NOT alter deployment architecture (Vercel)	✓
DOES NOT modify existing table schemas	✓
EXTENDS admin backoffice with new management sections	✓
EXTENDS public portal with new client-facing modules	✓
ADDS new database tables following existing conventions	✓
ADDS new API routes following existing patterns	✓

3. SCOPE OF CHANGE

3.1 In Scope

- Branded trainer profile PDF (1-page summary + full version), auto-generated on-demand.
- Client logo grid with categorisation, testimonials section, and case study cards.
- Live statistics counters (classes, hours, participants, clients).
- Intelligent training proposal generator with PDF output.
- Lead capture form with auto-email delivery (trainer profile PDF attachment).
- CRM-style lead pipeline in admin backoffice (Airtable or internal DB).
- WhatsApp direct contact button with pre-filled contextual message.
- Follow-up automation trigger (3-day no-response auto-follow-up).
- Google Analytics 4 integration + download/click event tracking.
- Admin BI dashboard: lead sources, page views, proposal requests, download counts.
- SEO metadata, Open Graph tags, sitemap, structured data (JSON-LD).
- Sticky bottom CTA bar on mobile (Request Training / Download Profile / WhatsApp).
- Page speed and Core Web Vitals optimisation.

3.2 Out of Scope (Future Phase)

- Student / client login portal.
- Payment gateway or course deposit collection.
- Multi-language (BM/EN) content switch.
- AI-powered auto-generated proposals without human review step.
- Real-time chat widget.

4. MODULE A — BRANDED PDF PROFILE DOWNLOAD

This module enables any website visitor or client to download a professionally designed, auto-generated PDF of Mohd Sulfri's trainer profile in two formats: a concise 1-page executive summary and a comprehensive multi-page full profile. Both formats are generated dynamically from live admin-managed portal data.

4.1 Functional Requirements

FR ID	Requirement
FR-100	The system must provide a 'Download Profile' button on the public homepage and profile page.
FR-101	The button must offer two download options: (1) 1-Page Executive Summary, (2) Full Trainer Profile.
FR-102	The PDF must be auto-generated server-side from live portal data at time of request — not a static file.
FR-103	The 1-Page Summary must include: name, headline, photo, top 5 certifications, core expertise areas, list of client organisations, and contact details.
FR-104	The Full Profile must include all content from the 1-Page Summary plus: detailed biography, complete certification list with badge images, full training expertise by domain, list of all client organisations with type labels, teaching experience summary, and downloadable resources list.
FR-105	PDF must use a branded layout: MSH colour palette, logo, professional typography, and consistent section styling.
FR-106	Each PDF must include a footer with: generation date, portal URL, trainer contact email, and 'Generated from sulfri-portal.vercel.app'.
FR-107	Admin must be able to toggle each section's inclusion in the PDF via admin settings.
FR-108	Admin must be able to update the profile photo, tagline, and contact details used in PDF output from the existing profile settings page.
FR-109	PDF generation must complete within 5 seconds on standard server load.
FR-110	The system must track: number of 1-page downloads, number of full profile downloads, date/time of each download event (stored for admin analytics).
FR-111	The system must NOT require a user account or email submission to download the PDF (fully anonymous public access).
FR-112	Admin must be able to download a preview copy of both PDF formats from the backoffice at any time.

5. MODULE B — CLIENT SHOWCASE ENHANCEMENTS

This module significantly strengthens the social proof and credibility signals on the public portal. It introduces testimonials, case study cards, animated live counters, and a structured client logo grid — all managed by admin and designed to build trust rapidly with prospective clients browsing the portal.

5.1 Functional Requirements

FR ID	Requirement
FR-113	Admin must be able to add a Testimonial entry with fields: Quote text, Name, Job Title, Organisation, Photo URL (optional), Rating (1–5 stars, optional), Visibility, Display order.
FR-114	Public homepage must display a testimonials carousel or grid showing all visible testimonials.
FR-115	Admin must be able to add a Case Study card with fields: Client name/label, Training topic, Number of participants, Duration, Outcome summary (max 200 chars), Date, Visibility, Display order.
FR-116	Public homepage must display a Case Studies section showing all visible case study cards.
FR-117	The homepage must display animated live counters for: Total Classes Completed, Total Training Hours Delivered, Total Participants Trained, Total Unique Client Organisations. Values must be sourced from admin-editable fields, not auto-computed.
FR-118	The existing client organisation grid must be enhanced to show: organisation logo image (admin-uploaded), organisation name, and type badge (Government / GLC / Corporate / Education).
FR-119	Admin must be able to upload a logo image for each client organisation entry.
FR-120	The client grid must support filter tabs by type (All / Government / GLC / Corporate / Education) — extending existing functionality.
FR-121	Admin must be able to feature up to 6 organisations as 'Featured Clients' displayed prominently above the full grid.
FR-122	Public profile must display a media/press section if admin has added at least one press entry (title, publication, URL, date).
FR-123	Admin must be able to add press/media entries with fields: Headline, Publication name, URL, Date, Visibility.
FR-124	All showcase sections must degrade gracefully — if no entries exist, the section is hidden (not shown as empty).

6. MODULE C — INTELLIGENT PROPOSAL GENERATOR

This module introduces a smart, guided form where a prospective client selects their organisation type, training topic area, group size, and preferred delivery mode. The system then auto-generates a branded, professional training proposal PDF — pre-populated with relevant trainer credentials, applicable training content, and pricing tiers. This dramatically reduces the time required for client canvassing.

6.1 Functional Requirements

FR ID	Requirement
FR-125	The system must provide a public-accessible proposal request form at /proposal.
FR-126	The form must collect: Contact name (required), Organisation name (required), Email (required), Phone (required), Industry/Sector (dropdown), Preferred Training Topic (derived from expertise nodes in CR-02), Group Size (radio: <20 / 20–50 / 50–100 / 100+), Preferred Delivery Mode (online / physical / hybrid), Preferred Timeline (dropdown: ASAP / 1 month / 3 months / Flexible), Additional notes (text, optional).
FR-127	Upon submission, the system must auto-generate a PDF training proposal using: submitted form data, relevant expertise node details (from CR-02 skill tree), matching Credly badges/certifications (from CR-01), trainer bio and contact from profile settings, and admin-configured pricing tiers.
FR-128	The generated proposal PDF must include sections: Cover page (client name, trainer name, date), Executive Overview, Proposed Training Programme (matched to topic selection), Trainer Profile Summary, Supporting Certifications, Delivery Options & Timeline, Investment / Pricing Overview, Next Steps & Contact.
FR-129	Admin must be able to define Pricing Tiers in the backoffice: Tier name, Price range (text field, e.g., 'RM 3,500 – RM 5,000 per day'), Description, Visibility.
FR-130	Admin must be able to customise the proposal cover page: background colour, logo placement, tagline.
FR-131	The generated proposal PDF must be emailed to both the submitter (client) and the admin (Sulfri) automatically upon generation.
FR-132	The system must store all proposal requests in the admin backoffice at /admin/proposals.
FR-133	Admin must be able to view, filter (by topic, date, status), and export all proposal requests.
FR-134	Admin must be able to update the status of a proposal request: New / Sent / Followed Up / Converted / Lost.
FR-135	Admin must be able to add internal notes to each proposal request.
FR-136	The proposal form must be linked from the homepage CTA ('Request a Training Proposal') and from the /expertise skill tree node detail panels (CR-02).
FR-137	The system must prevent duplicate proposal submissions from the same email within 48 hours (configurable). A user-friendly message must be shown.
FR-138	Admin must be able to regenerate and re-download the proposal PDF for any stored request at any time.

7. MODULE D — AUTOMATION, CRM & COMMUNICATION PIPELINE

This module converts the portal into an automated lead pipeline. Every lead — whether from the proposal form, event registration (CR-03), or a direct enquiry — is funnelled into a unified CRM view. Automated emails, follow-up triggers, and WhatsApp CTAs ensure no lead is lost after initial contact.

7.1 Functional Requirements

FR ID	Requirement
FR-139	All lead sources must be unified in a single admin lead pipeline view at /admin/pipeline: sources include Proposal Requests (CR-04 Module C), Event Registrations (CR-03), and Direct Enquiries (new contact form).
FR-140	A new public contact/enquiry form must be available on the homepage (modal or dedicated /contact page) with fields: Name, Email, Phone, Organisation, Message, Consent checkbox.
FR-141	Upon any new lead submission (proposal, event, or direct enquiry), the system must automatically send: (a) a confirmation/thank-you email to the lead including the 1-page trainer profile PDF attachment (Module A), and (b) a notification email to the admin (Sulfri).
FR-142	The confirmation email template must be admin-configurable: subject line, greeting, body text, signature, and attachment toggle.
FR-143	Admin must be able to configure a follow-up automation rule: if a lead's status remains 'New' for N days (configurable, default 3), the system must flag the lead for follow-up in the admin pipeline view.
FR-144	Admin must be able to send a follow-up email to a flagged lead directly from the pipeline view (one-click).
FR-145	The follow-up email template must be admin-configurable with merge fields: {lead_name}, {organisation}, {topic_interest}, {trainer_name}.
FR-146	Admin must be able to export the full lead pipeline to CSV at any time.
FR-147	A persistent WhatsApp CTA button must appear on all public pages linking to <code>wa.me/{admin_phone}?text={pre_filled_message}</code> .
FR-148	Admin must be able to configure the WhatsApp number and pre-filled message text from the backoffice.
FR-149	The pipeline view must show each lead's: source, submission date, status, organisation, topic interest, and last activity date.
FR-150	Admin must be able to manually update lead status: New / Contacted / Proposal Sent / Negotiating / Converted / Lost / Archived.
FR-151	Admin must be able to add internal notes and a follow-up reminder date to any lead.
FR-152	The system must display a dashboard widget on the admin homepage showing: total active leads, leads by status, leads by source, and conversion rate (Converted / Total).
FR-153	Email sending must use a transactional email service (e.g., Resend or SendGrid) configured via admin-supplied API key in environment variables.

FR ID	Requirement
FR-154	All outgoing emails must be logged (timestamp, recipient, template used, status: sent/failed) and viewable in admin at /admin/email-log.
FR-155	The system must handle email delivery failures gracefully: retry once, then flag as failed in the log without crashing the form submission.

8. MODULE E — ANALYTICS & ADMIN BUSINESS INTELLIGENCE

This module provides Sulfri with actionable insight into how the portal performs as a business tool: who is visiting, what they are downloading, which expertise topics attract the most proposal requests, and which events generate the most leads. All data is surfaced in a clean admin dashboard without requiring third-party logins.

8.1 Functional Requirements

FR ID	Requirement
FR-156	The admin homepage (/admin) must display a Business Intelligence (BI) Overview dashboard showing key metrics in a card layout.
FR-157	BI Dashboard must include the following metric cards: Total Page Views (30-day), Unique Visitors (30-day), PDF Downloads (1-page vs full, 30-day), Proposal Requests (30-day), Event Registrations (30-day), Lead Conversion Rate (%), WhatsApp CTA Clicks (30-day), Badge Wallet Views (30-day).
FR-158	Admin must be able to switch the BI Dashboard date range: Last 7 days / Last 30 days / Last 90 days / All time.
FR-159	The BI Dashboard must include a Top Topics chart showing which Training Expertise areas (from CR-02) received the most proposal requests.
FR-160	The BI Dashboard must include a Lead Sources chart showing breakdown of leads by source (Proposal Form / Event QR / Direct Enquiry).
FR-161	The BI Dashboard must include a Downloads Trend chart (line graph) showing daily PDF download volume over the selected period.
FR-162	Google Analytics 4 (GA4) must be integrated. The GA4 Measurement ID must be admin-configurable via a settings field (not hard-coded).
FR-163	The following events must be tracked via GA4 custom events: pdf_download (with parameter: type = summary full), proposal_form_submit, whatsapp_cta_click, badge_wallet_view, skill_tree_node_click, event_qr_scan (from CR-03).
FR-164	Admin must be able to view a Downloads Log table showing each PDF download event: date/time, PDF type, referrer page.
FR-165	Admin must be able to view a Proposal Requests analytics view: total requests by topic, by sector/industry, by month.
FR-166	The BI Dashboard data (internal metrics) must be stored in the portal's own database — GA4 is supplementary, not the sole data source.
FR-167	Admin must be able to export BI Dashboard summary data to CSV for external reporting.
FR-168	The BI Dashboard must clearly distinguish between internal tracked data (portal-native) and GA4 data (third-party) where both are shown.

9. MODULE F — SEO, PERFORMANCE & MOBILE UX

This module ensures the portal is discoverable on Google, loads fast on mobile networks, and provides a frictionless experience for clients browsing on smartphones — the primary device used by HR and procurement managers in Malaysia. It also introduces a sticky mobile CTA bar to minimise drop-off and maximise conversion.

9.1 Functional Requirements

FR ID	Requirement
FR-169	All public pages must include properly configured HTML meta tags: title (max 60 chars), description (max 160 chars), canonical URL. These must be admin-configurable per page via backoffice SEO settings.
FR-170	All public pages must include Open Graph tags (og:title, og:description, og:image, og:url) for correct social media preview rendering when links are shared on WhatsApp, LinkedIn, and Facebook.
FR-171	A JSON-LD structured data block (Person schema) must be injected into the homepage: trainer name, job title, organisation, skills, contact URL.
FR-172	A sitemap.xml must be auto-generated and accessible at /sitemap.xml listing all public pages.
FR-173	A robots.txt must be present at /robots.txt allowing all public pages and disallowing /admin routes.
FR-174	All public pages must achieve a Google PageSpeed Insights score of 85 or above on mobile (measured at deployment).
FR-175	Images must be served in next-gen formats (WebP) and must include width/height attributes to prevent Cumulative Layout Shift (CLS).
FR-176	A sticky bottom CTA bar must appear on mobile screens (max-width 768px) with three action buttons: 'Request Training' (links to /proposal), 'Download Profile' (triggers PDF download modal), 'WhatsApp' (links to wa.me CTA). The bar must not obstruct content and must include a dismiss button.
FR-177	The sticky CTA bar must be toggleable by the admin (show/hide) from the backoffice settings.
FR-178	All font loading must use font-display: swap to prevent invisible text during page load. External fonts must be preloaded.

10. DATABASE AMENDMENTS

The following new tables are added. All follow the naming conventions, audit field patterns (created_at, updated_at), and UUID primary key standard defined in SRS v1.0 and maintained across CR-01, CR-02, and CR-03.

10.1 testimonials

- id (PK, UUID)
- quote (TEXT, required)
- author_name (VARCHAR, required)
- author_title (VARCHAR, nullable)
- author_organisation (VARCHAR, nullable)
- photo_url (VARCHAR, nullable)
- rating (INT 1-5, nullable)
- visibility (PUBLIC | HIDDEN)
- display_order (INT)
- created_at (TIMESTAMP)
- updated_at (TIMESTAMP)

10.2 case_studies

- id (PK, UUID)
- client_label (VARCHAR, required)
- training_topic (VARCHAR, required)
- participant_count (INT, nullable)
- duration_text (VARCHAR, nullable)
- outcome_summary (TEXT, max 200 chars)
- study_date (DATE, nullable)
- visibility (PUBLIC | HIDDEN)
- display_order (INT)
- created_at (TIMESTAMP)
- updated_at (TIMESTAMP)

10.3 press_entries

- id (PK, UUID)
- headline (VARCHAR, required)
- publication (VARCHAR, required)
- url (VARCHAR, nullable)
- published_date (DATE, nullable)
- visibility (PUBLIC | HIDDEN)
- display_order (INT)
- created_at (TIMESTAMP)
- updated_at (TIMESTAMP)

10.4 proposal_requests

- id (PK, UUID)
- contact_name (VARCHAR, required)
- organisation (VARCHAR, required)
- email (VARCHAR, required)
- phone (VARCHAR, required)
- industry_sector (VARCHAR, nullable)
- topic_interest (VARCHAR, required)
- group_size (ENUM: <20 | 20-50 | 50-100 | 100+)
- delivery_mode (ENUM: online | physical | hybrid)
- preferred_timeline (VARCHAR, nullable)
- additional_notes (TEXT, nullable)
- generated_pdf_url (VARCHAR, nullable)
- status (ENUM: NEW | SENT | FOLLOWED_UP | CONVERTED | LOST)
- admin_notes (TEXT, nullable)
- consent_flag (BOOLEAN, required)
- submitted_at (TIMESTAMP)
- updated_at (TIMESTAMP)

10.5 pricing_tiers

- id (PK, UUID)
- tier_name (VARCHAR, required)
- price_range_text (VARCHAR, required)
- description (TEXT, nullable)
- visibility (PUBLIC | HIDDEN)
- display_order (INT)
- created_at (TIMESTAMP)
- updated_at (TIMESTAMP)

10.6 leads_pipeline

- id (PK, UUID)
- source (ENUM: PROPOSAL | EVENT | DIRECT_ENQUIRY)
- source_record_id (UUID, FK to respective source table)
- contact_name (VARCHAR)
- email (VARCHAR)
- organisation (VARCHAR, nullable)
- topic_interest (VARCHAR, nullable)
- status (ENUM: NEW | CONTACTED | PROPOSAL_SENT | NEGOTIATING | CONVERTED | LOST | ARCHIVED)
- follow_up_date (DATE, nullable)
- admin_notes (TEXT, nullable)
- last_activity_at (TIMESTAMP)
- created_at (TIMESTAMP)
- updated_at (TIMESTAMP)

10.7 email_log

- id (PK, UUID)
- recipient_email (VARCHAR, required)
- subject (VARCHAR)
- template_name (VARCHAR)
- status (ENUM: SENT | FAILED | RETRYING)
- sent_at (TIMESTAMP, nullable)
- error_message (TEXT, nullable)
- lead_id (UUID, FK to leads_pipeline, nullable)
- created_at (TIMESTAMP)

10.8 analytics_events

- id (PK, UUID)
- event_type (VARCHAR, required — e.g., pdf_download, whatsapp_click)
- event_data (JSONB, nullable — e.g., {type: 'summary'})
- referrer_url (VARCHAR, nullable)
- user_agent (VARCHAR, nullable)
- ip_hash (VARCHAR, nullable — hashed for privacy)
- created_at (TIMESTAMP)

10.9 portal_settings (extension)

- whatsapp_number (VARCHAR)
- whatsapp_prefill_message (TEXT)
- sticky_cta_enabled (BOOLEAN, default TRUE)
- ga4_measurement_id (VARCHAR)
- seo_homepage_title (VARCHAR)
- seo_homepage_description (TEXT)
- og_image_url (VARCHAR)
- proposal_duplicate_cooldown_hours (INT, default 48)
- follow_up_trigger_days (INT, default 3)
- email_confirmation_subject (VARCHAR)
- email_confirmation_body (TEXT)
- email_followup_subject (VARCHAR)
- email_followup_body (TEXT)
- pdf_include_certifications (BOOLEAN)
- pdf_include_expertise (BOOLEAN)
- pdf_include_clients (BOOLEAN)
- pdf_include_experience (BOOLEAN)

11. API AMENDMENTS

11.1 Public Endpoints

Endpoint	Method(s)	Description
/api/profile/download-pdf	GET	Generate and return branded PDF (param: type=summary/full)
/api/testimonials	GET	Return all PUBLIC testimonials
/api/case-studies	GET	Return all PUBLIC case studies
/api/press	GET	Return all PUBLIC press entries
/api/proposal/submit	POST	Submit proposal request form + trigger PDF generation + emails
/api/contact/submit	POST	Submit direct enquiry form
/api/analytics/track	POST	Log a client-side analytics event (pdf_download, whatsapp_click, etc.)
/api/settings/public	GET	Return public-safe settings (WhatsApp number, sticky CTA config, SEO meta)

11.2 Admin Endpoints (Protected — JWT/NextAuth Required)

Endpoint	Method(s)	Description
/api/admin/testimonials	GET, POST	List and create testimonials
/api/admin/testimonials/{id}	PATCH, DELETE	Edit or delete a testimonial
/api/admin/case-studies	GET, POST	List and create case studies
/api/admin/case-studies/{id}	PATCH, DELETE	Edit or delete a case study
/api/admin/press	GET, POST	List and create press entries
/api/admin/press/{id}	PATCH, DELETE	Edit or delete a press entry
/api/admin/proposals	GET	Paginated list of proposal requests with filters
/api/admin/proposals/{id}	GET, PATCH	View or update proposal status/notes
/api/admin/proposals/{id}/pdf	GET	Re-generate and download proposal PDF
/api/admin/proposals/export	GET	CSV export of all proposal requests
/api/admin/pipeline	GET	Unified lead pipeline view (all sources)
/api/admin/pipeline/{id}	PATCH	Update lead status, notes, follow-up date
/api/admin/pipeline/export	GET	CSV export of full pipeline

Endpoint	Method(s)	Description
/api/admin/pipeline/{id}/send-followup	POST	Trigger follow-up email to a lead
/api/admin/email-log	GET	Paginated email delivery log
/api/admin/pricing-tiers	GET, POST, PATCH	Manage pricing tiers for proposal generator
/api/admin/analytics/summary	GET	BI dashboard data (param: range=7d 30d 90d all)
/api/admin/analytics/export	GET	CSV export of analytics summary
/api/admin/settings	GET, PATCH	Manage all portal settings (SEO, WhatsApp, email templates, PDF toggles, GA4 ID)
/api/admin/profile/pdf-preview	GET	Download preview of both PDF profile formats

12. NON-FUNCTIONAL REQUIREMENTS

12.1 Security (Extension to SRS v1.0 Section 8.1)

- All form inputs (proposal, enquiry) must be server-side validated and sanitised.
- Rate limiting: POST /api/proposal/submit — max 5 per IP per hour; POST /api/contact/submit — max 10 per IP per hour.
- PDF generation endpoint must not expose internal file paths.
- Admin email API key (Resend/SendGrid) must be stored as environment variable only — never in database.
- Analytics event tracking must hash IP addresses before storage (PDPA compliance).
- GA4 Measurement ID must be stored in admin settings table and injected server-side — not hard-coded in client bundle.

12.2 Performance (Extension to SRS v1.0 Section 8.2)

- PDF generation (both formats) must complete server-side within 5 seconds.
- Proposal form page must load under 2 seconds on a standard 4G mobile connection.
- BI Dashboard analytics queries must return within 3 seconds for 90-day range.
- PDF download must be streamed — not held in memory — to prevent server timeout for large profiles.
- Analytics event tracking must be non-blocking (async, fire-and-forget pattern).

12.3 Compliance (Extension to SRS v1.0 Section 8.4)

- Consent checkbox mandatory on all lead-capture forms (proposal, enquiry, event — CR-03) per PDPA Malaysia.
- Admin must be able to permanently delete any individual lead record upon request (Right to Erasure).
- Email confirmation must reference trainer contact email for data deletion requests.
- IP address hashing in analytics_events table ensures no personally identifiable data is stored in analytics.
- PDF profile download (Module A) requires no personal data collection — fully anonymous.

12.4 Accessibility

- Sticky CTA bar must be keyboard-navigable and include ARIA labels.
- Proposal form must follow WCAG 2.1 AA: all fields labelled, errors associated via ARIA, keyboard navigable.
- BI Dashboard charts must include text summaries as accessibility fallback.

13. ACCEPTANCE CRITERIA

The feature set introduced in CR-04 is considered complete when ALL of the following criteria are verified:

Module A

1. Both PDF formats (1-page summary and full profile) are downloadable from the public portal.
2. PDF content is dynamically generated from live admin-managed data.
3. Download events are tracked and visible in the admin analytics section.
4. Admin can toggle PDF section inclusions and preview both formats from backoffice.

Module B

1. Testimonials, case study cards, and animated counters are visible on the public homepage.
2. Client organisation logos are displayed with type badges and filter tabs function correctly.
3. Press/media section appears only when at least one entry exists.
4. All showcase sections are hidden (not shown empty) when no content is published.

Module C

1. A prospective client can complete and submit the proposal request form.
2. A branded proposal PDF is auto-generated and emailed to both the client and admin.
3. Admin can view, filter, update status, and export all proposal requests.
4. Duplicate submissions from the same email within 48 hours are blocked with a user-friendly message.
5. Admin can regenerate the proposal PDF for any stored request.

Module D

1. All leads (proposals, events, enquiries) appear in the unified /admin/pipeline view.
2. Confirmation email with 1-page PDF attachment is sent to every new lead within 60 seconds.
3. Admin notification email is received for every new lead submission.
4. Leads exceeding the follow-up trigger window are flagged in the pipeline view.
5. Admin can send a follow-up email from the pipeline with one click.
6. WhatsApp CTA button is visible and functional on all public pages.
7. Email delivery log is accessible at /admin/email-log.

Module E

1. Admin BI Dashboard loads with key metrics for the selected date range.
2. Top Topics, Lead Sources, and Downloads Trend charts render correctly.
3. GA4 integration fires custom events for pdf_download, proposal_form_submit, and whatsapp_cta_click.
4. BI Dashboard summary data can be exported to CSV.

Module F

1. All public pages include title, description, canonical, and Open Graph meta tags.
2. Sitemap.xml and robots.txt are accessible and correctly configured.
3. Homepage achieves Google PageSpeed Insights mobile score of 85+.

4. Sticky bottom CTA bar is visible on mobile screens and dismissible.
5. Admin can enable/disable the sticky CTA bar from backoffice settings.

14. IMPACT ON DEPLOYMENT

Area	Change Required	Impact Level
Infrastructure	No additional servers or services	None
Hosting	Vercel — no model change required	None
Database	9 new tables + extension of portal_settings	Migration required
Frontend Routes	/proposal, /contact, /admin/pipeline, /admin/proposals, /admin/email-log, /admin/analytics, /sitemap.xml, /robots.txt	Route additions
Dependencies	PDF generation lib (Puppeteer or @react-pdf/renderer), QR lib (already added in CR-03), Email service SDK (Resend or SendGrid), GA4 (gtag.js)	Package additions
Environment Variables	EMAIL_API_KEY, GA4_MEASUREMENT_ID (moved to DB settings), PDF_SERVICE_URL (if external)	Config update
Build Time	PDF generation is server-side (API routes) — no build time impact	Minimal

15. VERSION CONTROL & CR LOG

Upon approval of CR-04, the main SRS version must be updated to v1.4. This CR must be merged into Sections 5 (Functional), 6 (Database), 7 (API), 8 (Non-Functional), and 10 (Acceptance Criteria). The CR log below must be maintained in the SRS Appendix.

CR ID	Target Version	Summary
CR-01	v1.1	Credly Badge Wallet & Skills Wallet Integration
CR-02	v1.2	Certificate-to-Training Expertise Linking + Visual Skill Tree
CR-03	v1.3	Event Lead Capture & YAYASAN PENERAJU Course Registration Module
CR-04	v1.4	Client-Facing Showcase Enhancement, Downloadable Profile, Automation & Business Intelligence Module

Version control recommendation: maintain a /docs/cr-log.md in the repository with each CR merged as a dated commit referencing the CR ID. Each deployment should be tagged with the target SRS version (e.g., git tag v1.4.0).

16. STRATEGIC VALUE STATEMENT

With the completion of CR-04, the Sulfri Trainer Portal transitions from a verified-credential showcase platform (CR-01 through CR-03) into a fully operational business development engine. The six modules introduced in this CR collectively deliver the following strategic outcomes:

**Zero-Friction Client
Canvassing**

Any prospective client who visits the portal can immediately download a professionally branded trainer profile PDF and submit a proposal request — all without picking up the phone.

Automated Lead Pipeline

Every event, every web visit, and every QR scan feeds automatically into a centralised pipeline. No leads are lost. Follow-up triggers ensure every opportunity is pursued within the configured window.

Proposal Velocity

The Intelligent Proposal Generator eliminates the hours typically spent drafting custom proposals for each client. A tailored, branded PDF is generated in seconds — increasing the volume of proposals Sulfri can send per day.

**Data-Driven Decision
Making**

The BI Dashboard surfaces which training topics attract the most interest, which events generate the most leads, and which lead sources convert best — enabling Sulfri to focus marketing and canvassing effort precisely.

Google Discoverability

Full SEO implementation ensures the portal surfaces in search results for relevant training queries in Malaysia, generating organic inbound leads without paid advertising.

**Professional Brand
Authority**

Testimonials, case studies, client logos, and a polished downloadable profile collectively build the level of social proof required for government and GLC procurement decisions.